METRIC
MIL-PRF-85045/22B
5 November 2021
SUPERSEDING
MIL-PRF-85045/22A
w/AMENDMENT 1
13 August 2018

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, EIGHTEEN FIBERS, STANDARD AND ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 2 (OFCC), APPLICATION B (SHIPBOARD), CABLE CLASS SM AND MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type: 2 (optical fiber cable component [OFCC])

Fiber optic cable class: MM (graded-index, glass core and glass cladding, multimode)

SM (dispersion-unshifted, glass core and glass cladding, single-mode)

DESIGN AND CONSTRUCTION:

Fiber: Class MM fibers shall be in accordance with MIL-PRF-49291/6.

Class SM fibers shall be in accordance with MIL-PRF-49291/7.

Buffer diameter: 900±50 micrometers (µm)

OFCC: Dimensions and configuration: See figure 1.

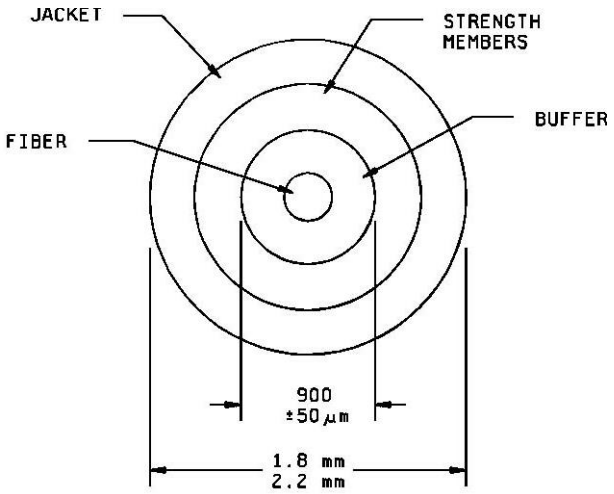


FIGURE 1. OFCC.

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Mass per unit length: <15 kilograms per kilometer (kg/km)

Tensile loading: >270 newtons (N)

Dynamic bend tensile load: 90 N minimum

Jacket material: The OFCC jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Cable jacket color: Shall be black or blue.

OFCC color:

Slate (MIL-PRF-49291/6 fiber).

Yellow (MIL-PRF-49291/7 fiber).

OFCC marking: Each OFCC shall be uniquely marked with a number between 1 and 18. The form of the marking shall be the printed spelling of the number, followed by a dash, followed by the printed Arabic numeral. The marking shall be applied and repeated every 0.10 meter (m) along the OFCC jacket.

OFCC marking for mixed fiber cables: OFCCs containing MIL-PRF-49291/6 fiber shall be marked with a number between 1 and the number representing the total number of MIL-PRF-4921/6 fibers in the cable. OFCCs containing MIL-PRF-49291/7 fiber shall be marked between the lowest number available, after all OFCCs containing MIL-PRF-49291/6 fiber have been marked, up to 18.

Short-term minimum bend diameter: Eight times the OFCC outer diameter.

Long-term minimum bend diameter: Sixteen times the OFCC outer diameter.

Finished cable:

Dimensions and configuration: See figure 2. Eighteen OFCC units shall be helically laid over the central member. OFCC 1 shall be located in the innermost ring of OFCCs. OFCCs shall be consecutively numbered starting with those in the innermost ring (1 through 6) and then those in the outer ring (7 through 18). OFCCs 1 and 7 shall be located along approximately the same radial line within the cable. The minimum outer jacket thickness shall be not less than 1.4 millimeters (mm).

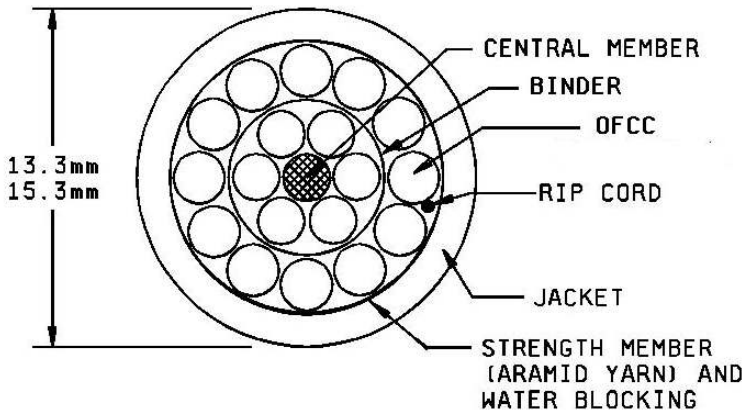


FIGURE 2. Eighteen-OFCC fiber optic cable.

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Number of fibers: 18 (one per OFCC)

Concentricity: >0.65

Mass per unit length: ≤215 kg/km

Jacket material: The overall jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Short-term minimum bend diameter: Eight times the cable outer diameter. (The short-term minimum bend diameter shall be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter: Sixteen times the cable outer diameter.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

PERFORMANCE REQUIREMENTS: The individual performance requirements shall be as specified in MIL-PRF-85045 and this specification sheet, including the requirements identified in table I.

Optical properties:

Maximum attenuation rate (Standard Performance):	4.5 decibels (dB)/km at 850±20 nanometers (nm) and 2.0 dB/km at 1,300±20 nm for class MM fiber.
	1.5 dB/km at 1,310±20 nm and 1,550±20 nm for class SM fiber.
Maximum attenuation rate (Enhanced Performance):	5.0 dB/km at 850±20 nm and 2.0 dB/km at 1,300±20 nm for class MM fiber.
	1.5 dB/km at 1,310±20 nm and 1,550±20 nm for class SM fiber.

For cables with radiation cross-linked jackets, the change in attenuation rate measurement may be made up to 30 days after cross-linking of the cable jacket.

Bandwidth: Fiber with a minimum bandwidth of 500 megahertz (MHz)-km at 1,300 nm and 160 MHz-km at 850 nm shall be used (multimode cables only).

Change in optical transmittance: Measurements shall be made at 1,300±20 nm. A minimum of eight fibers shall be monitored for change in optical transmittance. A minimum of two fibers shall be monitored in the inner ring of OFCCs. A minimum of six fibers shall be monitored in the outer ring of OFCCs. The fibers monitored in each ring shall be randomly selected, but different fibers shall be monitored in each test sample. For shock testing, only four fibers are required to be monitored (two in the inner ring and two in the outer ring of OFCCs).

Mechanical properties:

Tensile loading and elongation: Applicable; tensile loading ≥3,300 N.

Cyclic flexing: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable twist-bending: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable jacket tear strength: Applicable, except the cable jacket tear strength shall be not less than 35 N/centimeter (cm) of jacket thickness for standard performance cable.

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Hosing: Both low pressure and hydrostatic pressure are applicable.

Low pressure: Leakage shall be not greater than 60 milliliters (mL).

Hydrostatic: 2.1 megapascals (MPa)

Cable abrasion resistance:

Standard performance product: Applicable.

Enhanced performance product:

Cable scraping resistance: 750 cycles

Cable-to-cable abrasion: 500 cycles

Environmental properties:

Temperature range:

Operating:	-28 to 65 °C
Nonoperating:	-40 to 70 °C
Storage:	-40 to 70 °C

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Temperature humidity cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Standard performance product: The test shall be conducted as specified in MIL-PRF-85045.

Enhanced performance product: The test shall be conducted as specified in MIL-PRF-85045, except that the jacket material shall be tested at +175 °C for 4 hours.

Standard performance product: Exposure to automobile gasoline and tap water are not required and the test temperature for lubricating oil exposure shall be 73 to 77 °C. The tensile strength retention of the cable jacket material after exposure to hydraulic fluid shall be not less than 30 percent.

Enhanced performance product: Exposure to automobile gasoline and tap water is not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C), and lubricating oil (98 to 100 °C).

Chemical properties:

Halogen content: <0.2 percent

Cross-link verification: This test is applicable for cables with cross-linked jackets only. The test shall be conducted in accordance with ICEA T-28-562 and run at 200 °C. The test shall be in the qualification test sequence using a specimen cut from the sample used in the thermal shock test and in the Group III testing of the Group C conformance test sequence. The hot creep elongation shall not exceed 100 percent, and the hot creep set shall not exceed 10 percent.

Identification marking: For mixed fiber cables, the identification marking shall include the following: 18 Fiber - A by 62.5/125, B by 9/125, where A=number of 62.5/125 fibers and B=number of 9/125 fibers.

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Part or identifying number (PIN):

Standard performance:

M85045/22-01 (Multimode)

M85045/22-02 (Single-mode)

Enhanced performance:

M85045/22-01E (Multimode)

M85045/22-02E (Single-mode)

M85045/22-03E-XX XX (Mixed Fiber: XX Multimode fibers and XX Single-mode fibers)

- XX: 49291/6 fiber count
- XX: 49291/7 fiber count

Qualification and conformance inspection: See table I. This table specifies applicable requirements, qualification inspections, and conformance inspections per this specification sheet. The Requirement Paragraph and Test Paragraph columns within this table reference MIL-PRF-85045. The Cable Length column of this table applies only to the qualification inspection. A reduced sample set is allowed for conformance inspection; refer to MIL-PRF-85045 for conformance inspection sample size.

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TABLE I. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each 3/	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each 4/	A
II	Crosstalk	3.5.3	4.7.4.3	Three samples, 0.5 km each 4/	--
	Electromagnetic effects	3.7.16	4.7.6.16	One specimen, 6 m 4/	--
III	Temperature cycling	3.7.1	5/	Two samples, 0.5 km each 4/ (one on reel, one off)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each 6/	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each 6/	--
	Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	Two specimens, 8 m each 7/	B
	Cyclic flexing	3.6.4	4.7.5.4	Six specimens, 5 m each 8/ (two specimens at each temperature)	--
	Crush	3.6.5	4.7.5.5	Two specimens, 5 m each 7/	--
	Cable twist-bend	3.6.6	4.7.5.6	Six specimens, 5 m each 8/ (three specimens at each temperature)	C
	Impact	3.6.8	4.7.5.8	Two specimens, 5 m each 7/	C
	Temperature life (life aging)	5/	5/	Two specimens, 300 m each 7/	C
	Tensile loading and elongation	3.6.1	4.7.5.1	Two specimens, 150 m each 9/	C
	Operating tensile loading	3.6.1.1	4.7.5.1.1	Two specimens, 150 m each 10/	C
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each 7/	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each 7/	C

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length [1/, 2/]	Conformance inspection
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.49 km [4/] (on reel)	--
	Weathering	3.7.8	4.7.6.8	One specimen, 2 m [11/] and three material samples [12/]	--
	Fluid immersion	3.7.9	4.7.6.9	One specimen, 2 m [11/] and three material samples [12/] for each specified fluid	C
	Paint susceptibility	3.7.15	4.7.6.15	Two specimens, 2 m each [13/]	--
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m [11/]	--
	Shock	3.7.13	4.7.6.13	One specimen, 30 m [11/]	--
	Dynamic bend	3.6.2	4.7.5.2	One specimen, 150 m [14/]	B
	Hosing: low pressure	3.6.12.1	4.7.5.12.1	One specimen, 1.5 m [11/]	C
	Hosing: hydrostatic	3.6.12.2	4.7.5.12.2	One specimen, 2 m [11/]	C
	Radial compression (for application B only)	3.6.7	4.7.5.7	One specimen, 10 m [11/]	--
	Corner bend	3.6.9	4.7.5.9	Two specimens, 5 m each [13/]	--
	OFCC kink	[5/]	[5/]	Two specimens, 1 m each [13/]	--
	Cross-link verification	[5/]	[5/]	Two specimens, 1 m each [13/]	C
V	Dripping	3.6.13	4.7.5.13	One specimen, 30 cm [11/]	--
	Cable jacket tear strength	3.6.14	4.7.5.14	Five flat extruded jacket material strips [15/]	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	Five specimens [16/]	--
	Cable abrasion resistance	3.6.16	4.7.5.16	Four specimens, 2 m each [17/]	C
	Cable shrinkage	3.6.17	4.7.5.17	Three specimens, 0.5 m each [13/]	C
	Durability of identification	3.6.19	4.7.5.19	Three specimens, 2 m each [13/]	--
	Flame extinguishing and smoke generation	3.7.12.2	4.7.6.12.2	One specimen, 50 m [11/]	C
	Water absorption	3.7.14	4.7.6.14	Two specimens, extruded jacket material strips [16/]	--

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/	Conformance inspection
VI	Acid gas generation	3.8.1	4.8.1	One specimen, 1 m 18/	C
	Halogen content	3.8.2	4.8.2	One specimen, 1 m 18/	--
	Toxicity index	3.8.3	4.8.3	One specimen, 1 m 18/	C
	Smoke index	3.8.5	4.8.6	One specimen, 1 m 18/	C

NOTES:

1/ Tolerance on 0.5-km length is ±5 percent, provided that results are normalized to 1 km.
2/ Tolerance on shorter lengths is ±5 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2-m section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ As stated under the applicable tests in this specification sheet.
6/ The same samples as used in the temperature cycling test shall be used.
7/ A specimen cut from each sample used in the temperature cycling test shall be used.
8/ Three specimens cut from each sample used in the temperature cycling test shall be used.
9/ A specimen cut from each specimen used in the temperature life test shall be used.
10/ The same specimen as used in the tensile loading and elongation test shall be used.
11/ A specimen cut from the specimen used in the thermal shock test shall be used.
12/ Three flat extruded jacket material strips as specified in 4.7.5.15 shall be used in this test.
13/ Specimens cut from the specimen used in the thermal shock test shall be used.
14/ A specimen cut from the specimen used in the thermal shock test or one of the specimens used in the tensile loading and elongation test shall be used.
15/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition as the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 and obtained from flat extruded material that underwent the thermal shock test shall be used.
16/ Flat extruded jacket material as specified in 4.7.5.15 shall be used in this test.
17/ Specimen cut from the specimen used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion, and two specimens shall be used in the cable-to-cable abrasion testing.
18/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

Qualification by similarity to previously qualified cable:

Manufacturers who are qualified for M85045/17-01P or M85045/18-01P and M85045/20-01M, and whose multimode cable produced under this specification sheet passes the visual and mechanical and attenuation rate inspections specified herein, are qualified under this specification sheet for multimode enhanced performance cable (M85045/22-01E). This qualification by similarity is applicable only if the same OFCCs and same cable materials used in the previously qualified cables are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

Manufacturers who are qualified for M85045/17-01P or M85045/18-01P, and whose multimode cable produced under this specification sheet passes the tests and inspections specified in table II, are qualified under this specification sheet for multimode enhanced performance cable (M85045/22-01E). This qualification by similarity is applicable only if the same OFCCs and same cable materials used in the previously qualified cable are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

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TABLE II. Qualification and conformance by similarity to M85045/17-01P or M85045/18-01P.

Group	Inspection	Qualification inspection 1/	Conformance inspection 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Low temperature flexibility (cold bend)	X	B
	Impact (low temperature only)	X	C
IV	Hosing: low pressure	X	C
	Cable abrasion resistance	X	C
V	Flame extinguishing and smoke generation	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/17-01P or M85045/18-01P and are pursuing qualification for M85045/22-01E.

2/ Where manufacturers are qualified for M85045/17-01P or M85045/18-01P and M85045/22-01E and are performing conformance testing for M85045/22-01E.

Qualification by similarity – fiber type:

Manufacturers who are qualified under this specification sheet for multimode (M85045/22-01E) fiber cable and whose single-mode fiber cable passes the tests and inspections specified in table III are qualified under this specification sheet for single-mode (M85045/22-02E) fiber cable. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from a multimode fiber to a single-mode fiber. Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

Manufacturers who are qualified under this specification sheet for multimode fiber cable (M85045/22-01E) and single-mode fiber cable (M85045/22-02E) and whose mixed fiber cable passes the visual and mechanical and attenuation rate inspections specified herein are qualified under this specification for mixed fiber cable (M85045/22-03E). This qualification by similarity is applicable only if the same OFCCs and same cable materials used in the previously qualified cables are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

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TABLE III. Qualification and conformance by similarity (fiber).

Group	Inspection	Qualification inspection M85045/22-02E 1/	Conformance inspection M85045/22-02E 2/	Qualification inspection M85045/22-03E 3/	Conformance inspection M85045/22-03E 4/
I	Visual and mechanical	X	A	X	A
	Attenuation rate	X	A	X	A
III	Temperature cycling	X	C	--	--
	Temperature humidity cycling	X	C	--	--
	Storage temperature	X	--	--	--
	Cyclic flexing	X	--	--	--
	Crush	X	--	--	--
	Cable twist-bending	X	C	--	--
	Impact (low temperature only)	X	C	--	--
	Tensile loading and elongation	X	C	--	--
	Operating tensile loading	X	C	--	--
IV	Thermal shock	X	--	--	--
	Dynamic bend	X	B	--	--

NOTES:

1/ Where manufacturers are qualified for M85045/22-01E and are pursuing qualification for M85045/22-02E.

2/ Where manufacturers are qualified for M85045/22-01E and M85045/22-02E and are performing conformance testing for M85045/22-02E.

3/ Where manufacturers are qualified for M85045/22-01E and M85045/22-02E and are performing qualification testing for M85045/22-03E.

4/ Where manufacturers are qualified for M85045/22-01E, M85045/22-02E, and M85045/22-03E and are performing conformance testing for M85045/22-03E.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
NASA – NA

Review activities:
Army – AR, AV, MI
Navy – EC, YD
Air Force – 02, 19
DLA – CC

Preparing activity:
Navy – SH
(Project 6015-2020-007)

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